

Labor Input Responses to Ownership Changes: Evidence from U.S. Nursing Homes

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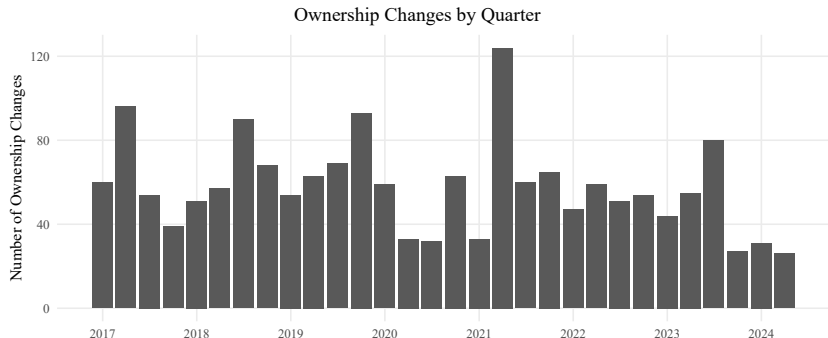
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- This second type of change does **not** alter the set of providers in a market.
- Instead, effects operate through **internal production decisions**.
- This type of change is understudied in the literature, and we examine it in **nursing homes**.

Ownership Changes in Nursing Homes Over Time (Quarterly)



Why Nursing Homes?

- 15,000 U.S. nursing homes serve over 4 million people annually
- 56% of retirement age adults will use a nursing home
- Labor is the main input into care
 - ▶ More staffing improves measured quality ([Lin, 2014](#)) and decreases deficiency citations ([Harrington et al., 2000](#))
 - ▶ Major cost
- Staffing can easily change so it may adjust following an ownership change

Research Question

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 - ▶ Do responses differ across staff type?
 - ▶ Do responses differ if a nursing home is part of a chain or not?
 - ▶ Do responses differ pre vs post-pandemic?

Nursing Staff Types

- **Nursing homes staff care using three main nursing roles:**
 - ▶ **Registered Nurses (RNs):** clinical assessment, care planning, meds, supervision
 - ▶ **Licensed Practical Nurses (LPNs):** routine clinical tasks, meds under RN oversight
 - ▶ **Certified Nursing Assistants (CNAs):** direct daily care
 - ▶ **Total:** RN + LPN + CNA

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 - ▶ **Total:** RN + LPN + CNA
- **Staffing measure: Hours Per Resident Day (HPRD)**
 - ▶ Standardizes staffing by resident census
 - ▶ Lets us compare facilities and months on a common scale
- Ownership change can affect both **levels** and **mix** which sets up cost-quality trade off.

Cost–Quality Trade Off

- Owners can raise quality through two channels:
 - ▶ **Quantity:** increasing total nursing hours (HPRD)
 - ▶ **Skill mix:** shifting hours toward more clinically trained staff
- **Both margins raise costs** because nursing labor is a major operating expense.
- **But short-run revenues are relatively fixed**, so staffing choices map into a **cost–quality tradeoff**.
- **Implication:** an ownership change that shifts objectives or constraints can move a facility along this tradeoff. ([Park and Werner, 2011](#); [Bowblis and Brunt, 2025](#))

Why Staffing Could Rise or Fall

- Ownership changes can shift **constraints** and **objectives** $\Rightarrow$ staffing effects are ambiguous.
- **Scenario 1: Cost pressure**
 - ▶ Debt service / tighter targets $\Rightarrow$ cut labor costs
- **Scenario 2: Quality/compliance push**
 - ▶ Invest in staffing to improve oversight / survey performance
- **Scenario 3: Reorganization**
 - ▶ Standardized staffing model shifts RN/LPN/CNA composition; total hours may be similar

Data Sources and Unit of Observation

- **Healthcare Cost Report Information System (HCRIS)**

- ▶ Annual cost report information
- ▶ Contains information about costs, revenue, financial data and dates of ownership change

- **Nursing Home Compare (NHC) Archive**

- ▶ Contains ownership information as well as facility characteristics

- **Payroll Based Journal (PBJ)**

- ▶ Payroll-reported nurse staffing hours by staff type
- ▶ Used to construct outcome variables

- **Facility-month panel, 48 contiguous states, Jan 2017 – Jun 2024**

Identifying Ownership Changes

- **Challenge:** ownership is often **layered**, so an administrative “change” may not reflect a true **transfer of economic control**.
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- We identify ownership changes using a **3-step process**.
- **Step 1:** HCRIS includes a variable indicating **whether and when** an ownership change occurred.
 - ▶ Useful for identifying **candidate events**, but the flag may be incomplete or capture legal restructuring rather than an economic change.
- Therefore, we have to use **NHC ownership shares** to verify and date whether control actually changes.

Ownership Level Challenge

- **NHC**

- We identify ownership changes at the **highest level of ownership present**

Ownership Level	Type	Owner Name	Share
<i>Direct ownership</i>			
Organization	LLC	BREDLEGS HOLDINGS LLC	25%
Organization	LLC	ROCK HILL ACQUISITION III LLC	75%
<i>Indirect ownership</i>			
Individual	Person	LALLY, THOMAS	25%
Individual	Person	LITT, ALAN	25%
Individual	Person	LITT, JONATHAN	25%
Individual	Person	STURM, JOSHUA	25%

Identifying Ownership Changes

- **Step 2: Validate & date using NHC**

- ▶ Track shares over time within the highest reported ownership level
- ▶ Define a change when **majority control (50%)** transfers between $t - 1$ and t
- ▶ We do not flag changes driven by name changes or within-family reclassification

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- **Example from the data:**

- ▶ *Before:*
 - ★ ESTES JAMES (84%)
 - ★ BURCHFIELD JOHN (10%)
 - ★ LEE CLAUDE (6%)
- ▶ *After:*
 - ★ JAMES N ESTES JR FAMILY DYNASTY TR NO 2 (50%)
 - ★ JENNIFER E AGEE FAMILY DYNASTY TR NO 2 (50%)
- ▶ We do **not** treat this as a meaningful ownership change

Ownership Change Agreement

- **Step 3: Validate between HCRIS and NHC.**
- We keep facilities only if:
 - ▶ **Never-treated:** 0 changes in **both** NHC and HCRIS, or
 - ▶ **Single change:** exactly 1 change in **each** dataset, within **6 months**.
- **Event month:** first month NHC shows a majority-control transfer.

Cross-tab of NHC & HCRIS

Cross-tab of Ownership Changes

Count of changes in NHC	2+	1,137 (8.5%)	1,377 (10.3%)	535 (4.0%)
	1	1,874 (14.0%)	1,589 (11.9%)	101 (0.8%)
	0	6,217 (46.6%)	499 (3.7%)	15 (0.1%)
		0	1	2+
		Count of changes in HCRIS		

Dependent Variables: Nurse Staffing (PBJ) and Sample

- We study four staffing measures:
 - ▶ RN, LPN, CNA, and total HPRD
- Outcome is **hours per resident day (HPRD)**: staffing intensity standardized by resident census

$$\text{HPRD}_{s,i,t} = \frac{\text{TotalHours}_{s,i,t}}{\text{ResidentDays}_{i,t}}$$

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- **CMS implausible staffing exclusions:**
 - ▶ RN HPRD = 0 **and** LPN HPRD = 0
 - ▶ Total HPRD < 1.5 or > 12
 - ▶ CNA HPRD > 5.25
- **Additional restrictions:** beds < 15; hospital-based facilities
- **Final sample:** 7,806 facilities; 632,231 facility-months; 1,589 treated ($\approx 20\%$)

Empirical Strategy: Event-Study Difference-in-Differences

- **Design:** staggered-adoption DiD around ownership transitions
([Goodman-Bacon, 2021](#))
- **Variation:** within-facility staffing changes relative to facilities not yet treated
- **Event study regression:**

Empirical Strategy: Event-Study Difference-in-Differences

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- **Event study regression:**

$$Y_{it} = \sum_{\tau=-24}^{24} \beta_{\tau} \mathbf{1}\{event_time_{it} = \tau\} \cdot Treat_i + X_{it}\gamma + \alpha_i + \lambda_t + \varepsilon_{it}.$$

- ▶ Y_{it} : staffing outcome in **HPRD**
- ▶ β_{τ} : effect on Y_{it} at event time τ
- ▶ X_{it} controls (beds, occupancy, payer mix, ownership type, chain, acuity)
- ▶ α_i facility FE; λ_t month FE
- ▶ SEs clustered at the facility and month level

Empirical Strategy: TWFE Difference-in-Differences

- **Goal:** summarize the **average** staffing change after an ownership transition
- **TWFE DiD regression:**

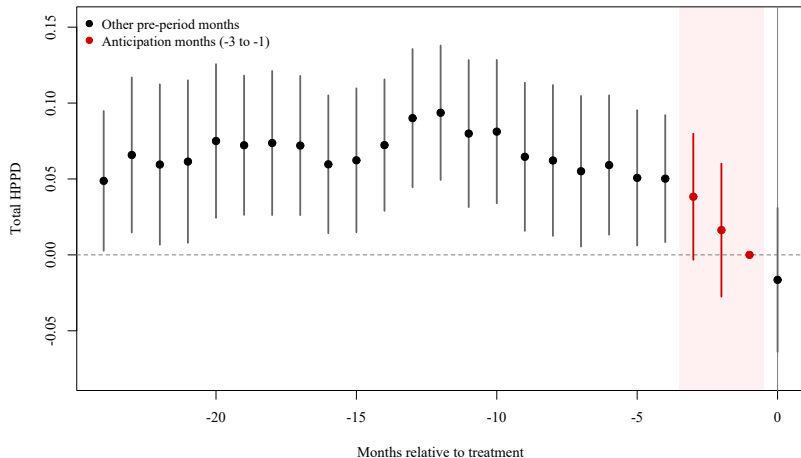
$$Y_{it} = \beta Post_{it} + X_{it}\gamma + \alpha_i + \lambda_t + \varepsilon_{it}$$

- ▶ $Post_{it} = 1$ in all months after a change
- ▶ β : **average post-change effect** on staffing Y_{it}
- ▶ Same fixed effects and controls as the event-study specification
- ▶ SEs clustered at the facility and month level

Identifying Assumptions

- **Parallel trends**
- **Staffing may change in anticipation of a formal ownership change**
 - ▶ Changes are not instantaneous: negotiations and operational handoffs precede the recorded month
 - ▶ Staffing can adjust during planning (turnover, hiring freezes, renegotiated contracts)
 - ▶ Recorded dates may reflect administrative/legal timing, not first operational change
- **Implication:** months just before $\tau = 0$ may violate parallel trends.

Evidence of violation in the Pre-Period



Main Specification: Donut Event Study + Pre-trend Test

- **Donut event study:** mitigate this influence by excluding months just before the recorded transition.
- **Implementation**
 - ▶ Exclude $\tau \in \{-3, -2, -1\}$
 - ▶ Use $\tau = -4$ as the reference period

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- **Implementation**
 - ▶ Exclude $\tau \in \{-3, -2, -1\}$
 - ▶ Use $\tau = -4$ as the reference period
- **Parallel Trends:** joint Wald test of pre-period leads

$$H_0 : \beta_\tau = 0 \quad \forall \tau < 0$$

- **Takeaway:** full pre-window shows evidence of pre-trends; the donut specification is more consistent with parallel trends.

Parallel Trends Tests (Joint Wald) — Total Staffing

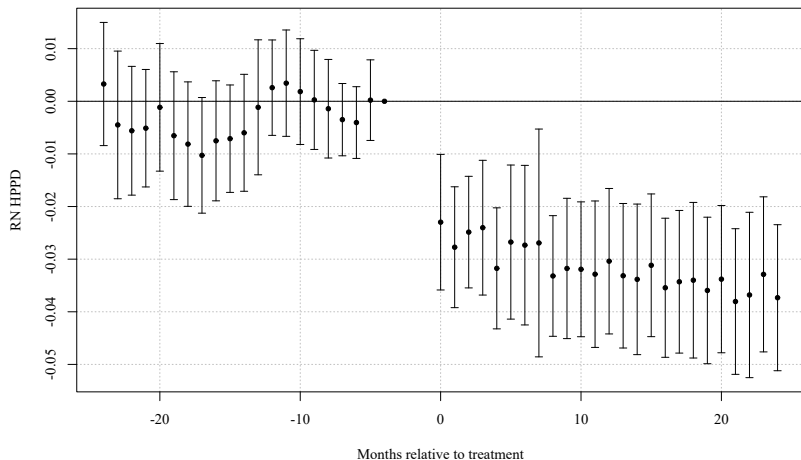
Table: Pre-trend Wald Test p-values (Total Staffing)

	Total (Levels)	Total (Logs)
2 Year Full Pre-Window	0.0006	0.0012
2 Year Window with Donut	0.2046	0.3268
1 Year Window with Donut	0.6387	0.6102

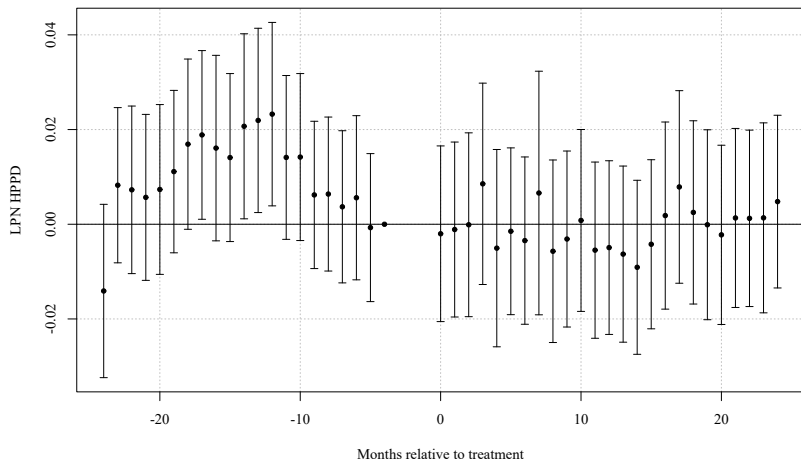
Notes: Joint Wald χ^2 test of $H_0 : \beta_\tau = 0$ for all pre-treatment coefficients. Cells report p-value.

Donut excludes $\tau \in \{-3, -2, -1\}$; reference period is $\tau = -4$.

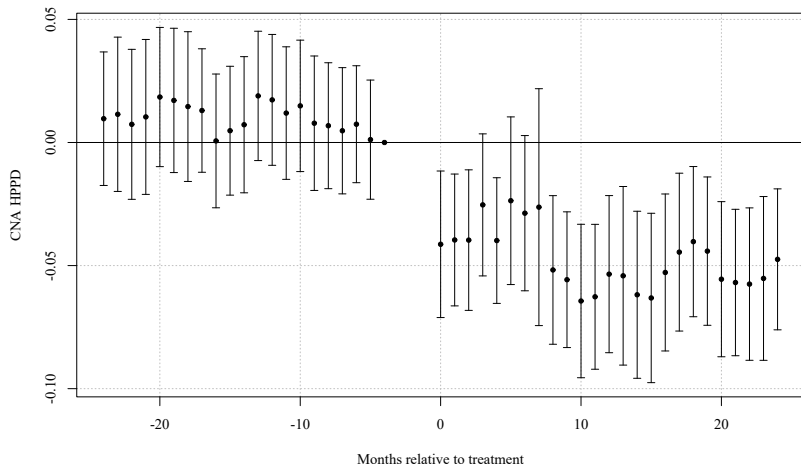
Event Study Results (RN HPRD)



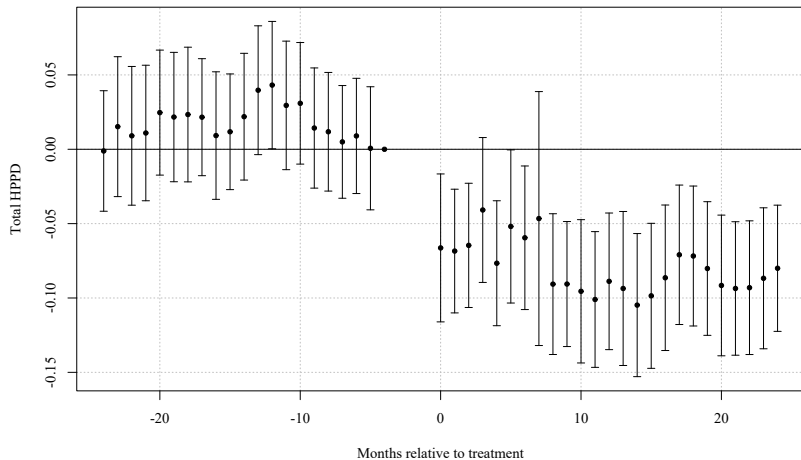
Event Study Results (LPN HPRD)



Event Study Results (CNA HPRD)



Event Study Results (Total Nursing HPRD)



TWFE DiD — Baseline (RN)

	RN	LPN	CNA	Total
HPRD	−0.032*** (0.005)	0.000 (0.006)	−0.054*** (0.009)	−0.086*** (0.013)
Log(HPRD)	−0.092*** (0.016)	0.003 (0.009)	−0.028*** (0.005)	−0.026*** (0.004)

Notes: Each cell reports the coefficient on *post* with two-way clustered SEs (facility and month) in parentheses. $N = 628,107$

TWFE DiD — Baseline (LPN)

	RN	LPN	CNA	Total
HPRD	-0.032*** (0.005)	0.000 (0.006)	-0.054*** (0.009)	-0.086*** (0.013)
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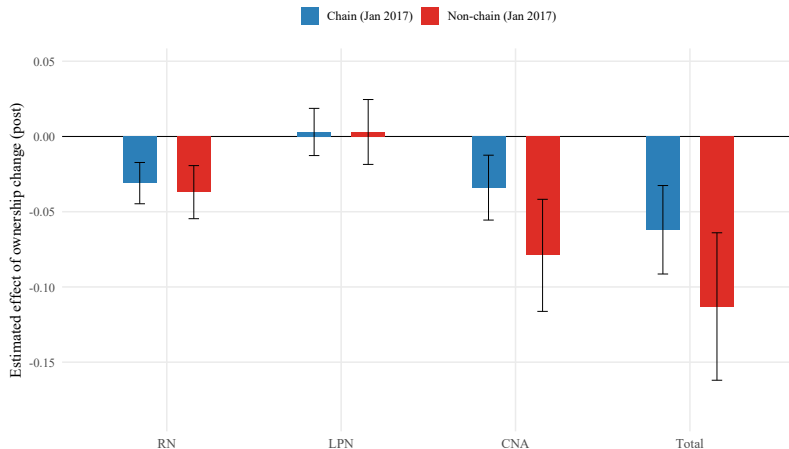
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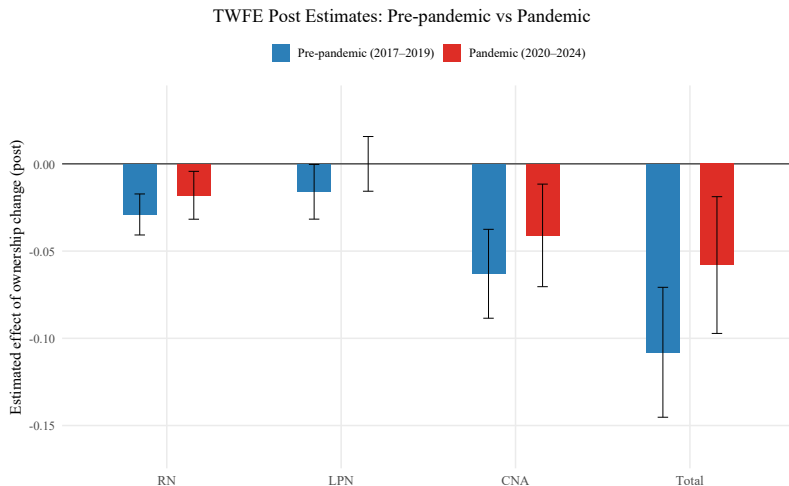
Heterogeneity: Chain vs Non-Chain Facilities

TWFE Post Estimates: Chain vs Non-chain



Staffing declines are larger for non-chain facilities, nearly double for CNA and Total.

Heterogeneity: Pre vs Post-Pandemic



Large pre-pandemic declines, and post-pandemic effects are almost cut in half.

Robustness Checks

- **Stacked Difference-in-Difference**
- **Smaller and wider Donut Hole**
- **Gaps in our Data.**

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- **Effects are heterogeneous across settings.**
 - ▶ Larger declines **pre-pandemic** vs. **pandemic period**
 - ▶ Larger declines for **non-chain** vs. **chain** facilities

Discussion

- **Key finding:** RN, CNA, and Total hours fall after ownership change; **LPN hours do not.**
- **Why might LPN hours not decline?**
 - ▶ **Minimum licensed coverage:** facilities must staff enough **RN+LPN** for clinical/compliance needs.
 - ▶ **Cheaper licensed nurse:** cutting **RN** while keeping **LPN** stable reduces labor costs.
 - ▶ **Substitution:** LPNs can absorb some routine clinical / direct-care tasks $\Rightarrow$ **skill-mix shift.**

(e.g., Lin 2014; Harrington et al. 2000; Bowlblis & Ghattas 2017)

Discussion

- **Question:** Are our findings economically **meaningful**?
- **Baseline effects (minutes per resident-day):**
 - ▶ **RN:** $-0.032 \text{ HPRD} \approx -1.9 \text{ minutes}$
 - ▶ **CNA:** $-0.054 \text{ HPRD} \approx -3.2 \text{ minutes}$
 - ▶ **Total:** $-0.086 \text{ HPRD} \approx -5.2 \text{ minutes}$
- [Lin \(2014\)](#) finds that a **0.3 HPRD** increase in RN staffing (≈ 18 minutes) improves deficiencies by 16% .
 - ▶ Our results would translate to a 1.6% increase in deficiencies.

Discussion

- **Heterogeneous effects**

- ▶ Staffing declines are **larger in non-chain facilities**
- ▶ Staffing declines are **smaller after the Covid-19 pandemic**

- **Why are declines larger in non-chain facilities?**

- ▶ Smaller labor pools
- ▶ Weaker hiring and less standardization
- ▶ Less ability to absorb shocks without cutting variable costs
- ▶ Ownership changes may be more destabilizing without chain-level management systems

- **Why are declines smaller after Covid?**

- ▶ Widespread shortages make staffing reductions harder to implement
- ▶ Greater scrutiny of staffing/quality raises the cost of cutting hours
- ▶ Facilities closer to staffing constraints have less room to reduce hours further
- ▶ Owners may prioritize stabilizing operations rather than restructuring staffing during crisis periods

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 - ▶ Linking our findings to quality metrics and resident outcomes

Thank you

Questions?

References I

- Bowblis, J. R. and Brunt, C. S. (2025). Nursing home staffing expenditures and levels are impaired by high medicaid payer-mix. *Journal of the American Medical Directors Association*, 26(8):105723.
- Goodman-Bacon, A. (2021). Difference-in-differences with variation in treatment timing. *Journal of Econometrics*, 225(2):254–277.
- Gupta, A., Howell, S. T., Yannelis, C., and Gupta, A. (2021). Owner incentives and performance in healthcare: Private equity investment in nursing homes. Working Paper 28474, National Bureau of Economic Research.
- Harrington, C., Zimmerman, D., Karon, S. L., Robinson, J., and Beutel, P. (2000). Nursing home staffing and its relationship to deficiencies. *The Journals of Gerontology: Series B*, 55(5):S278–S287.
- Holmes, J. S. (1996). The effects of ownership and ownership change on nursing home industry costs. *Health Services Research*, 31(3):327–346.

References II

- Lin, H. (2014). Revisiting the relationship between nurse staffing and quality of care in nursing homes: An instrumental variables approach. *Journal of Health Economics*, 37:13–24.
- Park, J. and Werner, R. M. (2011). Changes in the relationship between nursing home financial performance and quality of care under public reporting. *Health Economics*, 20(7):783–801.